

Applications of Interactive Style Transfer to Virtual Gallery

ABSTRACT

Since the outbreak of the epidemic in recent years, many events have been held online to avoid spreading illness, such as the virtual gallery. Therefore, we propose an interactive style transfer virtual gallery system. Users can perform style transfer on 2D objects and 3D objects, where 2D objects include pictures and paper jams, and 3D objects include frames and 3D artworks. The results show that it not only makes the viewing process more interesting but also deepens the impression of the artwork through the process of choosing different styles.

Index Terms: Style Transfer—Virtual Gallery; Human-centered computing—Human computer interaction(HCI)—User studies;

1 INTRODUCTION

In the current virtual galleries, curators and viewers can only do limited manipulation of the artworks, such as basic zooming and moving. Considering that the artworks in the virtual gallery are all digitalized, we can use this information to develop new interactive experiences further.

We compare the results of different style transfer methods and select two of them to integrate into our system based on our considerations. In our virtual gallery, users can perform style transfer on 2D objects and 3D objects through the user interface, where 2D objects include pictures and paper jams, and 3D objects include frames and 3D artworks.

2 RELATED WORK

2.1 Image Style Transfer

The image style transfer can be traced back as far as the algorithm proposed by Gatys et al. [1] They demonstrated that the high-dimensional features extracted from the image through the DNN model can represent the overall structure of an image, and the correlation between features can be used to capture the style of an image. They design loss function based on those two findings and optimize the original image by multiple iterations. Thus the resulting image can maintain the structure of the content image and have the characteristics of the style image.

The previous arbitrary style transfer approaches are holistic transformations that dismissed local style patterns. Park et al. [6] introduced self-attention mechanism to learn the semantic correlation between the content features and the style features for generating more local style details. Liu et al. [4] indicated that the low-level patterns of images also have an impact on the style transfer result.

2.2 Video Style Transfer

Compared with image style transfer, the goal of video style transfer is to focus on not only generating pleasing stylization results but also on considering the consistency of transferred results between adjacent frames of the video so as to avoid flickering problem. Li et al. [3] proposed a linear transformation model, mainly to learn a transformation matrix to replace whitening and coloring transforms. Since the linear transformation can preserve content affinity at test time, the stylized videos are more stable and consistent. Deng

et al. [7] calculate per-channel correlation across content and style representations. The generated video satisfies the constraint that the flicker of the video is not perceived by the human eye.

2.3 3D Style Transfer

3D style transfer requires more consideration of the three-dimensional information of 3D objects than image style transfer. Otherwise, transferred results will have various visual artifacts. Kato et al. [2] proposed to use of a 3D rendering process to render an image of the 3D object from a random angle as the input of the CNN model to perform style transfer. Mordvintsev et al. [5] also use the 3D rendering process to turn 3D objects into 2D RGB images, but unlike the method of Kato et al. [2], they do not modify the geometry of the 3D object. The texture of the 3D object is optimized by backpropagating the gradient of the loss function proposed by Gatys et al [1]. In addition, not only transfer the textures of 3D objects, but some works focus on transferring geometric style between 3D objects.

3 SYSTEM IMPLEMENTATION

The application is developed with Unity 3D 2020.3.15f2. User predominantly experiences virtual gallery from a first-person perspective with the camera at a fixed height and capable of rotating 360°. User interaction is done with a mouse and keyboard. The user interacts with objects and manipulates the user interface through a mouse and the user is enabled to use WASD keys or arrow keys to move in the virtual environment.

3.1 Object Type

In the virtual gallery, a wide range of 2D and 3D artworks are displayed in this space for viewers to appreciate. In addition, the different styles of the frame will bring different emotions to viewers. We choose the typical Chinese frame as the basic frame in our virtual gallery. The paper jam in it plays the role of embellishment and accent, which will also have a specific influence on the overall artwork.

In our virtual gallery, users can perform style transfer on 2D objects and 3D objects, where 2D objects include pictures and paper jams, and 3D objects include frames and 3D artworks (Figure 1).

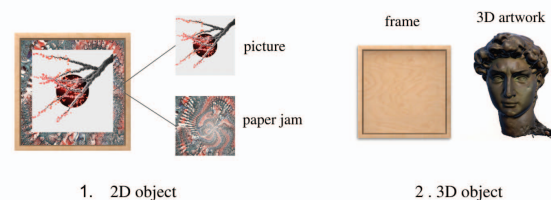


Figure 1: Two types of objects.

3.2 Transferred Results

Fig. 2a shows the results of performed style transfer on the frame using the picture. Compared to the original plain frame, we observe

that the stylized frames seem to be designed especially for this artwork because it has similar color tone and style elements to the picture. Fig. 2b shows the result of using the picture to perform style transfer on the paper jam. We observed that the transferred paper jam can present the effect of extension. Fig. 2e and Fig. 2f show the transferred results of the two 3D artworks, respectively. We observe that the transferred results are more vivid and artistic due to the addition of stylistic textures.



Figure 2: (a-c) Results using picture. (d-f) Results using arbitrary style.

4 SYSTEM EVALUATION

4.1 User Study Design

In order to evaluate the effectiveness of our system, we conduct a within-subjects user study based on measuring user satisfaction towards transferred results and evaluating system usability by questionnaires and user interviews. A total of fourteen 2D artworks are hung on the wall in the scene, each containing a picture, paper jam, and frame (Fig. 3). All frames have the same wood texture.



Figure 3: Screenshots of the scene.

4.2 Tasks

In order to clearly identify the participant's perceptions of different objects performing style transfer, we divided the task into four parts according to the transformed objects, including frame transformation, paper jam transformation, arbitrary transformation, and 3D artwork transformation, where the arbitrary transformation task meant that the user could perform style transfer on the picture, paper jam, and frame at will. We asked participants to use their sense of aesthetics to transform the objects and compare the overall feeling of the transferred objects to the original ones using the questionnaire.

5 DISCUSSION

In this section, we summarized the positive feedback and negative feedback on performing style transfer on different objects based on the experimental results and user interviews.

Frame Transformation: Frames are the object participants like to perform style transfer most. According to the participants' feedback, we observed that most participants indicated that the color and style patterns of the transferred frame were consistent with the picture. Therefore, the overall result after style transfer is more harmonious and artistic. Some participants also mentioned that choosing a suit-

able style to transfer frames will not only increase the richness of the picture but also highlight the content of the picture.

Paper Jam Transformation: Most participants who preferred to stylize the paper jam selected a style image that is very different from the original picture. They said it could improve the contrast with the picture. Most participants indicated that if the transferred result has more patterns or strong colors would make the artwork look messy and reduce the overall aesthetics.

3D Transformation: In the preference questionnaire, 3D artwork was ranked lower than the other objects. The participants who liked the transferred result said that the style transfer could add rich colors to the original monotonous 3D artwork and enhance the overall look.

Visual Effect of Transformation Speeds: Regarding the speed of the transformation process, participants who like slow transformation the most said they could watch the transformation process more clearly. Furthermore, remain the transferred result until the user leaves the area because they consider that this method will not interrupt the process of appreciating the artwork and can also increase the naturalness of the experience.

6 CONCLUSION

The results show that incorporating style transfer into the virtual gallery can make the experience more interesting and deepen the user's impression of the artwork. Some participants gave new stories to the artworks when viewing the transferred artworks. Thus, we conclude that style transfer can effectively increase the connection between the viewer and the artworks. From our experimental results, most participants thought the transferred result of the frame was better than the original. We found that the frame transformation was more effective than other objects in improving the overall aesthetics of the artwork. In addition, participants thought the transferred result using the picture was slightly better than using their selected style. On the other hand, the paper jam transformation and 3D artwork transformation with current methods have little effect on enhancing the aesthetics. We need to develop or find other methods to improve.

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REFERENCES

- [1] L. A. Gatys, A. S. Ecker, and M. Bethge. Image style transfer using convolutional neural networks. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 2414–2423, 2016. doi: 10.1109/CVPR.2016.265
- [2] H. Kato, Y. Ushiku, and T. Harada. Neural 3d mesh renderer. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2018.
- [3] X. Li, S. Liu, J. Kautz, and M.-H. Yang. Learning linear transformations for fast image and video style transfer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2019.
- [4] S. Liu, T. Lin, D. He, F. Li, M. Wang, X. Li, Z. Sun, Q. Li, and E. Ding. Adaattn: Revisit attention mechanism in arbitrary neural style transfer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 6649–6658, October 2021.
- [5] A. Mordvintsev, N. Pezzotti, L. Schubert, and C. Olah. Differentiable image parameterizations. *Distill*, 3, 07 2018. doi: 10.23915/distill.00012
- [6] D. Y. Park and K. H. Lee. Arbitrary style transfer with style-attentional networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2019.
- [7] M. Ruder, A. Dosovitskiy, and T. Brox. Artistic style transfer for videos. In B. Rosenhahn and B. Andres, eds., *Pattern Recognition*, pp. 26–36. Springer International Publishing, Cham, 2016.